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Optimization of Sports Training Intensity Based on Fuzzy Control

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Abstract

Optimizing sports training intensity is essential for enhancing athletic performance while minimizing the risks of overtraining and injury. Traditional methods often lack the adaptability needed to accommodate an athlete's dynamic and physiological responses. To address this, the study introduces the Fuzzy Control-based Training Optimization Framework (FC-TOF), an intelligent and adaptive approach to managing training intensity. The FC-TOF integrates fuzzy control techniques with real time physiological data, such as heart rate variability, perceived exertion, and training volume. These inputs are processed through a fuzzy logic system to provide personalized, dynamic training intensity adjustments based on individual performance goals and physical conditions. The FC-TOF has been validated using historical athlete data and tested on participants from various sports disciplines. The implementation of FC-TOF exhibited significant enhancements in training results. Athletes exhibited increased endurance, superior peak performance measures, reduced injury rates, and more effective recovery management than traditional techniques. The technology promoted improved adherence to training schedules by offering personalized, data-informed recommendations. This study highlights the efficacy of FC-TOF as a useful tool for improving sports training intensity. The study provides a dependable basis for personalized training methodologies, improving athletic performance while mitigating training-related hazards.

Keywords Sports training optimization · Fuzzy control · Training intensity management · Adaptive framework · Athletic performance · Injury prevention · Nonlinear physiological responses · Intelligent systems

1 Preliminary Overview

Optimization of sports training intensity is key to improving performance. Traditional strategies like fixed training programs and general progressions have improved performance but often fail to meet human physiology's complexity and variability [1]. The resulting stiffness can cause underperformance, overtraining, or injury, reducing training program effectiveness [2]. Genetic predisposition, recovery capability, surrounding factors, and psychological states affect the responses to training stimuli, making it difficult to create a universal training protocol [3].



Recent advances in wearable electronics, physiological monitoring devices, and AI have transformed training optimization [4]. Heart rate variability and blood lactate devices can reveal athletes' preparation, weariness, and recuperation. Real-world usage of the complex and big data is difficult [5]. Sports science uses fuzzy control systems more since it manages complex conditions and uncertainty. Fuzzy logic, a mathematical framework for uncertain data, fits nonlinear and ever-changing human physiology [6]. Fuzzy systems can imitate expert coaching with language factors and rule-based decision-making. The study can adjust to an athlete's state in real time, ensuring accurate and responsive training recommendations and avoiding overtraining or undertraining [7].

Sports adaptation is complicated, yet typical training frameworks emphasize volume, Intensity, and frequency without considering individual athlete needs [8]. Human reactions to training inputs are rarely linear, consequently, the same stimulus may have falling returns or negative effects. Poor intensity control can cause overtraining syndrome, persistent exhaustion, performance decline, and injury risk, whereas insufficient Intensity can cause underpreparedness and performance stagnation [9]. Despite being evident, experienced coaches' management may not work for all athlete demographics. Smart systems process enormous physiological data and create tailored training recommendations using data analytics, machine learning, and fuzzy control [10]. These systems can grow and adapt to individual differences, shifting training paradigms.

Continuous athlete monitoring, physiological adaptation mechanisms, and customized training responses are needed to optimise sports training intensity. Athlete monitoring often combines objective data like HRV and training volume with subjective input like RPE to assess internal and external load. Recovery, cumulative effort, and tolerance levels affect physiological adaptation, which improves, plateaus, or diminishes. Due to genetic, environmental, and psychological factors, athletes respond differently to identical training stimuli, making customized adjustment vital. The research introduces the FC-TOF framework, an adaptive, data-driven approach that helps coaches and athletes manage training loads while accounting for physiological and performance variability.

The novelty of this study lies in the FC-TOF framework's integration of fuzzy inference and LSTM networks, incorporating multi-dimensional physiological and subjective metrics with temporal analysis. This adaptive, data-driven approach enables personalized, context-aware training load recommendations, outperforming traditional static models in responsiveness, precision, and athlete-specific optimization.

The novel Fuzzy Control-based Training Optimization Framework (FC-TOF) manages training intensity using fuzzy logic and deep learning [11]. Real-time physiological data changes training intensity for athletes. The structure improves monitoring and decision-making [12]. This FC-TOF hybrid design uses the Sugeno Fuzzy Inference System (FIS) and Multivariate LSTM networks [13]. The Sugeno FIS [14] models expert training adaptive knowledge with if-then rules to interpret imprecise and ambiguous data. LSTM networks [15] capture temporal dependencies, making them perfect for assessing physiological response patterns across time. The FC-TOF predicts athlete adaptability and adapts training recommendations based on historical data.

The main objectives are.

- To create a Fuzzy-Control Training Optimization Framework (FC-TOF) that combines Sugeno Fuzzy Inference System or FIS and Multivariate Long Short Term Memory or LSTM networks to provide precise and real-time adjustments to sports training intensity based on physiological data.
- To develop an adaptive, data-driven training optimization system (FC-TOF) capable of providing personalized recommendations that can assist athletes and coaches in managing training load, supporting endurance development, peak performance preparation, and recovery planning, while minimizing overtraining and injury risks.
- To address athlete's dynamic and nonlinear physiological responses by leveraging fuzzy logic and deep learning techniques, ensuring effective long-term planning and real-time adaptability in sports training.

The studies are summarized here. The second section gives a complete review of the body of research and method of approach. Section 3 specifies the processing techniques, study design, and approach. Section 4 addresses the analysis's results. Future activities and the conclusion are found in Sect. 5.

2 Literature Review

An [16] emphasized the role of fuzzy decision support systems in enhancing the effectiveness of sports fitness training programs. The approach applied fuzzy decision-making models to improve the quality of program design and achieved greater efficiency in training delivery. While the method demonstrated notable improvements in decision accuracy, its application was limited to specific training scenarios. The study highlighted the need for extending such approaches to a wider range of training domains and broader athletic contexts.

Fang et al. [17] developed a wavelet recursive fuzzy neural network to enable real-time modulation of physical training intensity. This method successfully delivered adaptive and optimized intensity adjustments during training, offering real-time responsiveness. However, the computational requirements for real-time processing were relatively high, which may affect deployment in certain scenarios. Further work was recommended to enhance scalability for various sports and training types.

Qu [18] presented a fuzzy neural network-based approach for regulating workout intensity. The method improved both the precision and timeliness of intensity adjustments, ensuring more accurate monitoring. Its performance, however, relied heavily on high-quality input data, and challenges remained in maintaining reliability under dynamic and noisy conditions. The study suggested addressing these limitations to improve system robustness.

Huang et al. [19] applied fuzzy classification models to classify and analyze activities in sports training sessions. The system achieved accurate activity categorization and provided valuable analytical insights to enhance training quality. Despite its strengths, generalization to other sports beyond the tested types was limited, indicating opportunities to adapt the method for multi-sport or varied activity settings.

Lu and Wang [5] designed fuzzy control algorithms to intelligently monitor Taijiquan exercises. The approach enabled real-time monitoring and performance evaluation for practitioners. While effective within Taijiquan, the method's applicability was restricted to this martial art, suggesting the need for adaptation to other combat sports and exercise forms.

Chen et al. [20] proposed a fuzzy neural network model to improve management in college physical education programs. The system enhanced resource allocation and program quality, demonstrating the value of intelligent decision support in educational sports contexts. Implementation, however, required specialized expertise, which could hinder widespread adoption. The authors noted the potential for extending the model to other institutions and course structures.

Dziomdziora and Taibi [21] combined fuzzy logic and machine learning to optimize running pace and training distance. This integration improved the precision of pace and distance adjustments, resulting in better performance outcomes. The method's dependency on the accuracy of the machine learning component was a limitation, and further validation across broader running conditions and diverse demographics was advised.

Wan [22] utilized fuzzy clustering techniques and artificial intelligence models to design aerobic physical education training programs. The approach improved flexibility and adaptability in program design. However, its high computational requirements could restrict implementation in resource-limited environments. Long-term performance impacts on athletes were identified as an important area for future study.

Fouad et al. [23] created a heart rate monitoring system for athletes using fuzzy clustering. The system provided real-time, accurate monitoring, contributing to improved safety and training effectiveness. Dependence on specific hardware and sensors was a limitation, prompting the need for more cost-effective and scalable alternatives.

Song [24] applied multi-objective optimization algorithms to develop balanced and efficient sports programs for college students. The method optimized program design to cater to diverse student needs. Nevertheless, the approach required extensive data for accurate optimization, and future work could focus on expanding its applicability to a wider range of sports and activities.

The research methods and results of different authors are shown in Table 1.

Table 1 Review of existing literature

Authors	Objective	Methods	Findings	Advantages	Disadvantages	Research Gap
An [16]	Improve training programs' effectiveness in sports fitness using fuzzy decision support systems.	Applied fuzzy decision-making models	Enhanced decision-making for training program design	Improved training efficiency and effectiveness	Limited to specific training scenarios	Applicability of the approach to other domains or broader training programs
Fang et al. [17]	Physical training intensity modulation in real time using wavelet recursive fuzzy neural networks.	Developed a fuzzy neural network with wavelet transforms	Achieved effective real-time intensity adjustments	Real-time adaptability and optimization	Computational complexity in real-time scenarios	Scalability to different training types or sports
Qu [18]	Control workout intensity with a fuzzy neural network	Implemented fuzzy neural networks	Improved accuracy and timeliness in training intensity adjustments	Enhanced precision in training intensity monitoring	Requires high-quality input data	Addressing system limitations in dynamic environments
Huang et al. [19]	Classify and analyze activities during sports training using a fuzzy model	Used fuzzy classification models	Accurate activity classification and detailed training analysis	Improved training session insights	Limited generalization to other types of sports training	Expanding to multi-sport scenarios or varied activities
Lu and Wang [20]	Intelligent Taijiquan exercise monitoring using fuzzy control theory	Designed fuzzy control algorithms	Effective monitoring of Taijiquan exercises	Provides real-time monitoring and evaluation	Limited applicability to Taijiquan	Adapting fuzzy logic to other martial arts or exercise forms
Chen et al. [21]	Fuzzy neural networks improve college physical education management.	Developed a fuzzy neural network model	Enhanced management of college-level physical education programs	Optimized resource allocation and program quality	Requires specialized expertise for implementation	Extending the approach to diverse educational institutions and courses
Dziomdziora and Taibi [22]	Fuzzy logic and machine learning adjust Running pace and training distance.	Combined fuzzy logic with machine learning	Achieved better pace and distance optimization	Integration of ML and fuzzy systems for enhanced outcomes	Dependency on machine learning model accuracy	Broader validation across varied running conditions and demographics
Wan [23]	Apply AI and fuzzy logic to aerobic physical education training	Utilized fuzzy clustering techniques and AI models	Improved aerobic training program design and implementation	Integration of AI and fuzzy systems for greater flexibility	High computational requirements	Examining long-term impacts on athletes' performance
Fouad et al. [24]	Create an athlete heart rate monitoring system utilizing fuzzy clustering.	Created a fuzzy clustering-based heart rate monitoring system	Achieved real-time, accurate heart rate monitoring	Improved athlete safety and training efficiency	Hardware and sensor dependency	Developing cost-effective and scalable solutions
Song [25]	Create college sports programs utilizing multi-objective optimization.	Applied multi-objective optimization algorithms	Designed efficient and balanced sports programs for college students	Optimized sports programs for diverse student needs	Requires extensive data for accurate optimization	Expanding methodology to cover a wider variety of sports programs

3 Proposed Methodology

A Fuzzy Control-based Training Optimization Framework efficiently manages sports training intensity. Fuzzy control and real time physiological data, including Heart Rate Variability(HRV), rate of perceived exertion, and training volume, enable personalized, dynamic training intensity changes. The goals include optimization, endurance, peak performance, recovery, and overtraining/injury risk reduction. The key features are Sugeno Fuzzy Inference System (FIS), Multivariate LSTM Network, adaptive training optimization, and personalized suggestions. The FC-TOF adapts training intensity to fit individual needs and performance goals based on real-time physiological data. Fuzzy logic and deep learning power the framework's intelligent, adaptive, and personalized training optimization.

Figure 1 demonstrates the way the comprehensive Fuzzy Control-Based Training Optimization Framework (FC-TOF) optimizes sports training intensity using intelligent data processing and adaptive control. HRV, RPE, Training Volume, Recovery Score, and Performance Score are needed. The core preprocessing unit collects and normalizes raw input data, creates standardized data formats for the Multivariate LSTM network and Fuzzy

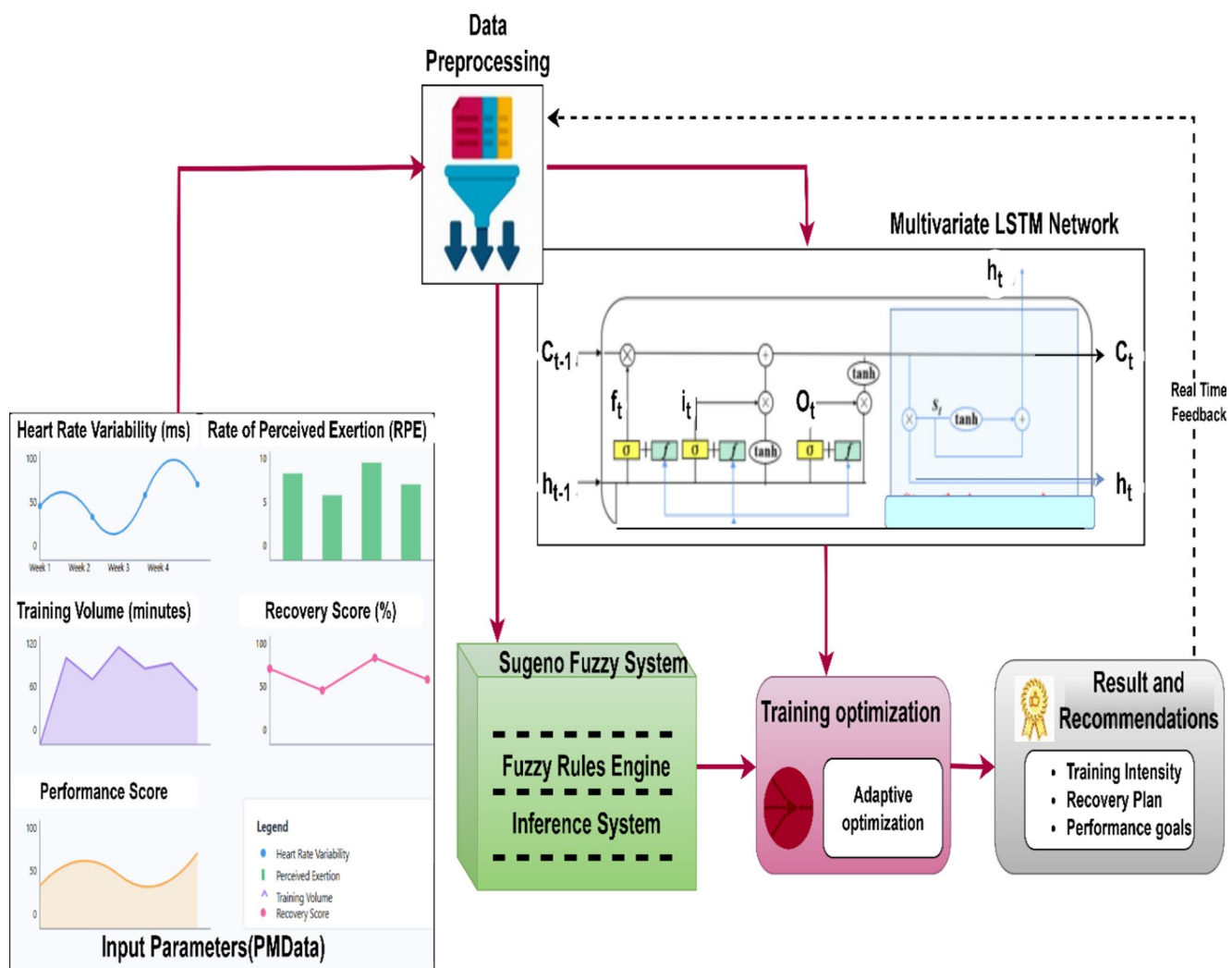


Fig. 1 Proposed FC-TOF

System, and provides a feedback loop for system adaption. A Fuzzy Rules Engine processes expert information, an Inference System decides, and the green fuzzy system uses linguistic variables and fuzzy logic.

Training Optimization uses adaptive optimization techniques, real-time adjustments, and direct recommendations. Output delivers training intensity guidelines, recuperation plan suggestions, performance goals and targets, and actionable insights for athletes and coaches. Primary data flow, feedback loops from findings to preprocessing, integration points between components, and processing stage separation are shown in Fig. 1. Dual Processing Approach, adaptive learning, data usage, real-time processing, and personalization are new system features. The FC-TOF framework aims to provide precise, real-time training intensity adjustments, use immediate and historical data for decision-making, offer personalized recommendations based on individual responses, balance performance enhancement and injury prevention, and adapt to dynamic physiological responses. A comprehensive sports training management system combines advanced technologies (LSTM and Fuzzy Logic) with practical training parameters.

3.1 Data Preprocessing

Data preparation is necessary for accurate, dependable, and informative physiological data analysis. The preprocessing pipeline optimizes athlete physiological data processing for fuzzy control-based training intensity variations. The FC-TOF adjusts training intensity using physiological data. It measures heart rate variability (HRV) and rate of perceived exertion (RPE), an athlete's subjective assessment of training effort. Duration and Intensity determine training volume (Fig. 2).

For the research, physiological inputs are characterized as follows: Heart Rate Variability (HRV) is expressed in milliseconds (ms) and obtained through the time-domain root mean square of successive differences (RMSSD); Rate of Perceived Exertion (RPE) is scored on the Borg CR-10 scale, ranging from 0 (no effort) to 10 (maximal effort); and Training Volume (TV) is computed as the product of session duration (in minutes) and training intensity (rated on a 1–5 scale), with the result represented in arbitrary training units (ATU). These parameter definitions establish a consistent reference for all athletes and supply uniform inputs to both the fuzzy inference mechanism and the LSTM network.

Several data pretreatment pipeline phases ensure appropriate inputs for the Fuzzy-Control Training Optimization Framework (FC-TOF). This process includes acquiring raw data, handling missing data, finding and removing outliers, normalizing, integrating, standardizing, and encoding. Raw physiological data including HRV, subjective exertion, and training volume is collected and processed for completeness. Mean imputation and box plots fill in missing data and highlight outliers to prevent biased analyses. Equation 1 is used in the fourth stage of normalization to compare all variables.

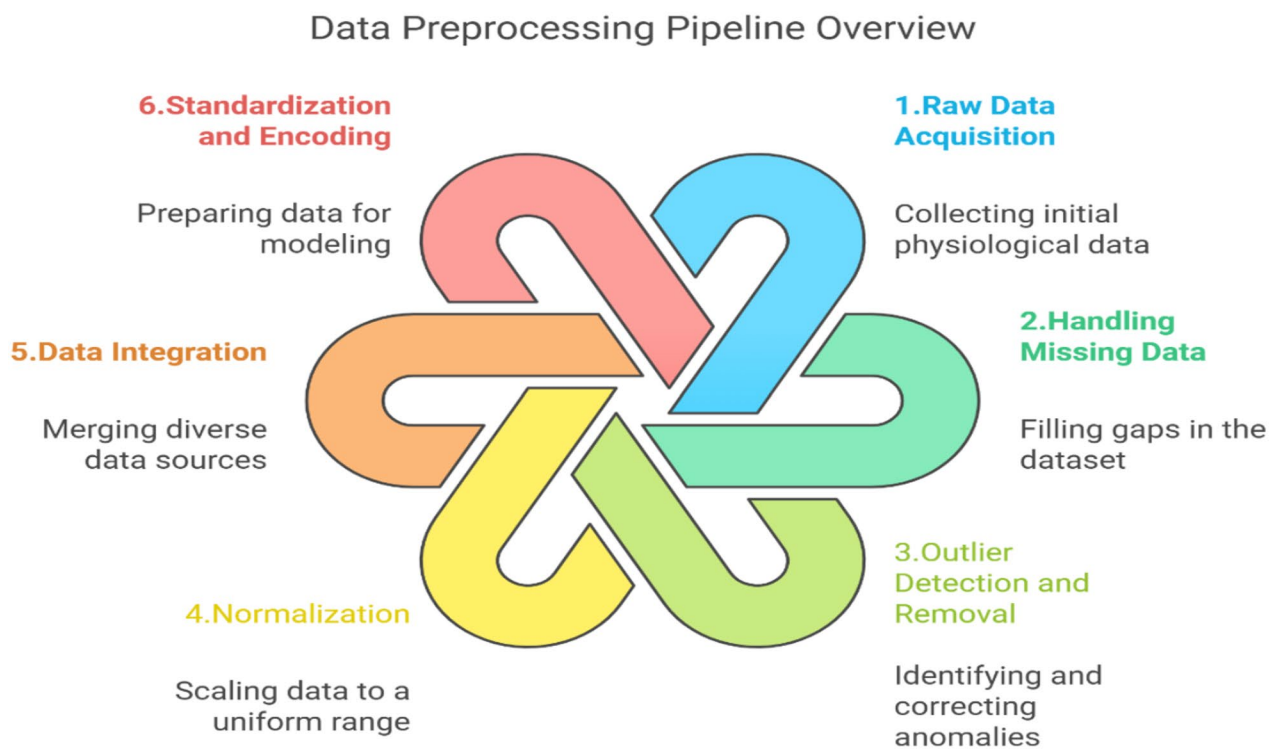


Fig. 2 Overview of data preprocessing

$$X_{norm} = \frac{(X - X_{min})}{(X_{max} - X_{min})} \quad (1)$$

In Eq. (1), X_{norm} is represented as the value that is normalized of X , X_{min} is represented as the min value of X in the dataset, and X_{max} is represented by the max value of X in the dataset. By bringing all variables to the same scale, normalization removes the potential for bias due to data magnitude differences. Data integration is creating a single dataset from several sources, such as physiological measurements taken in real time, expert opinions, and performance records. Standardization and encoding ensure it works with the deep learning models and fuzzy control system. The FC-TOF framework relies on a clean, normalized, and integrated dataset as input, guaranteeing precise, individual, and adaptable adjustments to training intensity.

In Table 2, The data consists of columns for each athlete, including their unique identifier, heart rate, subjective effort, training duration, intensity level, and training volume. The data is standardized using a formula to get the heart rate, subjective effort, and training volume. Due to its logging of heart rate data, its inclusion or reasonable estimation of training time and Intensity from workout sessions, and its derivation of training volume from duration and Intensity, this data is consistent with PMData. The subjective effort is presumably generated from performance feedback. Variables with a scale from 0 to 1 undergo the usual preprocessing steps. This aligns with PMData since it gives a thorough picture of how the athlete did during workouts.

Heart rate measurements are recorded continuously throughout the training session and subsequently segmented into three distinct phases: (i) Warm-up phase—initial low-intensity activity intended to prepare the cardiovascular and musculoskeletal systems; (ii) Main activity phase—consisting of steady-state or interval-based exercises where intensity varies according to the prescribed training load; and (iii) Recovery phase—post-exercise period where heart rate progressively returns toward baseline. For the data in Table 2, the heart rate values represent the average measurements taken during the main activity phase, ensuring consistency across all athletes and training sessions.

The original dataset [25] was collected using wearable fitness trackers (Fitbit Versa 2.0) synchronized with the PMSys sports application, complemented by self-reported performance feedback through structured Google Forms. Each training session's physiological and subjective metrics were recorded in real time and stored in a secure database. For example, consider Athlete ID 003 from Table 2: heart rate was captured at 1-second intervals during the main activity phase of a 75-minute high-intensity interval training session. The athlete reported a Rate of Perceived Exertion (RPE) of 8 immediately post-session via the Borg CR-10 scale. Training duration and recorded intensity level (5) were used to calculate Training Volume as $75 \times 5 = 375$ min. This raw data point was then normalized to a scale of 0–1 (HR = 0.50, RPE = 0.8, Volume = 1.0) before integration into the FC-TOF framework for model input.

The FC-TOF framework considers not only the individual values of HRV, RPE, and Training Volume but also their temporal evolution and interrelationships. For instance, a pattern of declining HRV accompanied by increasing RPE over consecutive sessions may indicate early signs of overreaching, prompting the system to recommend reduced intensity or extended recovery.

Table 2 Sample preprocessed data from PMData

Athlete ID	Heart Rate (bpm)	Subjective Effort (1–10)	Training Duration (min)	Training Intensity (1–5)	Training Volume (min)	Normalized HR	Normalized Effort	Normalized Volume
001	75	6	60	4	240	0.35	0.6	0.8
002	90	4	45	3	135	0.55	0.4	0.45
003	85	8	75	5	375	0.50	0.8	1.0

3.2 Sugeno Fuzzy Inference System

The Sugeno Fuzzy Inference System uses fuzzy logic principles to model the link between physiological inputs and training intensity outputs. It uses a rule base to evaluate clean inputs and calculate a defuzzified output for adaptive and real-time decision-making.

3.2.1 Fuzzification

The fuzzification process uses membership functions to transform clean inputs into fuzzy sets. This procedure ranks the input variables according to their membership, which includes HRV, RPE, and training volume. The triangular membership function is defined as Eq. (2):

$$\mu_{HRV}(x) = \begin{cases} 1, & x = a \\ \frac{c-x}{c-b}, & b < x \leq c \\ 0, & x > c \end{cases} \quad (2)$$

Equation 2, a , b , and c are the membership function's lower, peak, and upper bounds. $\mu_{HRV}(x)$ is represented by the membership degree for HRV at value x . The raw time-series data undergoes fuzzification, a critical step where crisp input values are transformed into fuzzy sets. This transformation involves assigning membership degrees to linguistic variables such as "Low HRV," "High RPE," or "Moderate TV". Fuzzy sets.

model the imprecision and uncertainty in physiological data using membership functions (such as triangular or Gaussian). "Low," "Normal," and "High" are some possible fuzzifications of HRV readings; overlapping membership functions would represent the subtle changes between these states.

The output of the fuzzification process is a set of fuzzy variables $\{A_{i1}, A_{i2}, \dots, A_{in}\}$, where each A_{ij} represents the membership degree of the j th variable in the i th category. These fuzzy variables form the basis for subsequent rule-based inference.

In Fig. 3, the Fuzzy-Control Training Optimization Framework (FC-TOF) uses fuzzy logic and deep learning to revolutionize sports training intensity management. It generates tailored, dynamic, and effective training recommendations using real-time physiological data and adaptive fuzzy rules. The approach uses data-generated fuzzy rules to link input variables to training intensity modifications. Dynamically generated from past athlete data, the fuzzy rule base captures complicated physiological variable interactions. Rule simplification reduces redundancy and improves computing performance. The framework employs the matching fuzzy rule process to find the most relevant rules for a testing sample and fuzzy rule extrapolation to generate new rules when new conditions arise.

Defuzzification converts fuzzy outputs into actionable values in the last stage. Fuzzy logic and Multivariate LSTM networks improve long-term adaptability. To complement the fuzzy rule-based system, LSTM networks examine past time-series data to capture temporal dependencies and forecast future physiological states. The PMData Sports Logging Dataset validated the FC-TOF, showing better endurance, recovery management, injury rates, and training schedule adherence. The FC-TOF bridges data-driven methods and tailored athletic care, laying the groundwork for intelligent sports training system study.

Table 3 shows fuzzy control system input variable membership function settings. HRV, RPE, and Training Volume use triangle membership functions with lower bound (a), peak (b), and upper bound (c) for exact fuzzy rule generation for dynamic training intensity optimization.

3.2.2 Fuzzy Rule Base Construction

The fuzzy rule base is the core of the FC-TOF. The fuzzy rule contains data-generated rules that transmit the input variables to training intensity modifications. Each rule monitors the structure IF (Condition1) AND (Condition2) THEN (Action), where the conditions are created on the fuzzified inputs, and the actions specify the corresponding intensity adjustment. For example:

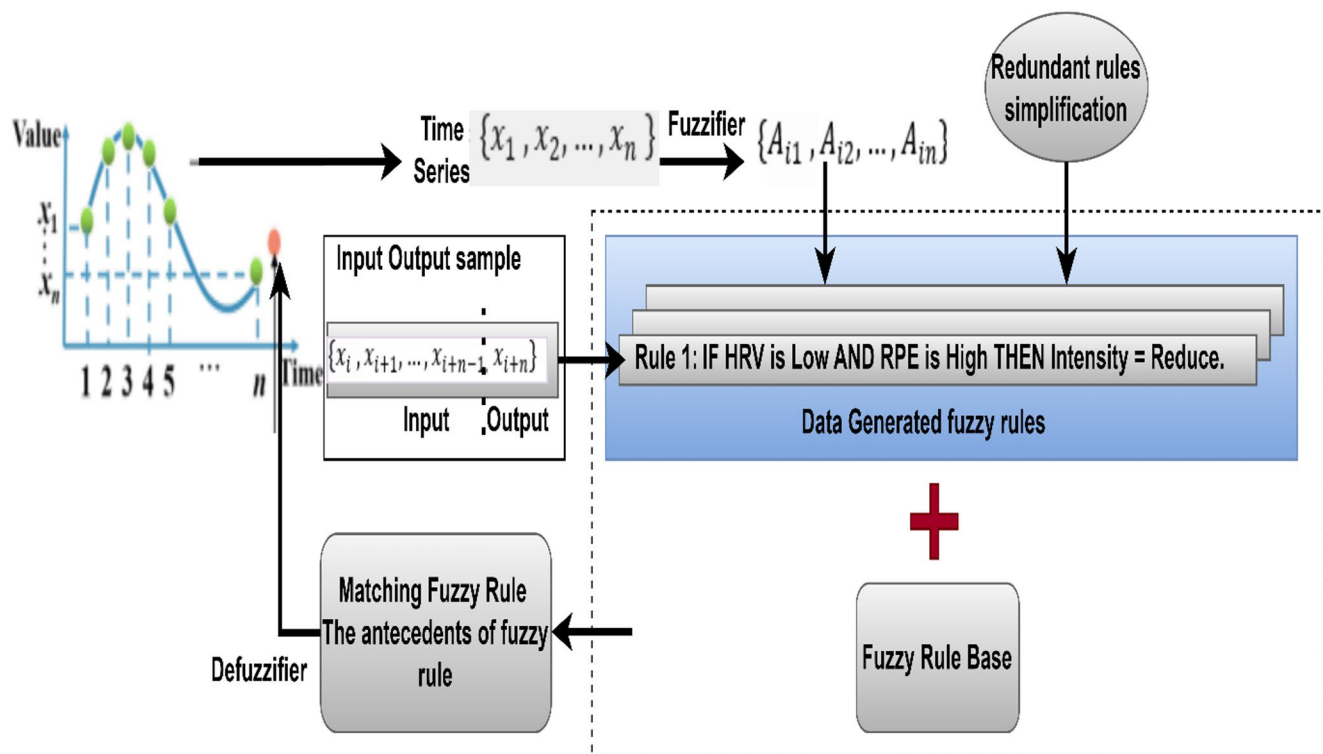


Fig. 3 Fuzzy inference system for sports training intensity optimization

Table 3 Membership function parameters

Input variable	Membership function	Lower bound (a)	Peak(b)	Upper bound(c)
HRV	Triangular	30	60	90
RPE	Triangular	5	15	25
Training Volume	Triangular	20	50	80

Table 4 Fuzzy rule table

Rule	HRV	RPE	Training Volume	Intensity
1	<i>Low</i>	<i>Low</i>	<i>Low</i>	<i>Increase</i>
2	<i>Low</i>	<i>Medium</i>	<i>Medium</i>	<i>Maintain</i>
3	<i>High</i>	<i>High</i>	<i>High</i>	<i>Reduce</i>

- **Rule 1:** If HRV is Low AND RPE is High, THEN Intensity=Reduce.
- **Rule 2:** If HRV is Normal AND RPE is Moderate, THEN Intensity=Maintain.

Based on athlete history, the fuzzy rule foundation is dynamic. It shows complicated physiological relationships, enabling exact training intensity recommendations.

Table 4 presents a fuzzy rule set that guides the adjustment of training intensity based on three input variables: HRV, RPE, and Training Volume. Each rule specifies a unique combination of these inputs and corresponding output action. When HRV, RPE, and Training Volume are low, the Intensity is increased to boost training output. If HRV is low and RPE and Training Volume are medium, the Intensity is maintained for balance.

3.2.3 Inference and Defuzzification

The Sugeno Fuzzy Inference System (FIS) services a *min-max* approach to regulate the notice strength of each rule. For a given rule i , the firing strength (w_i) is calculated using the following Eq. (3):

$$w_i = \min(\mu_{HRV}, \mu_{RPE}, \mu_{Volume}) \quad (3)$$

In Eq. (3), w_i represents the shot strength of rule i , while μ_{HRV} , μ_{RPE} , μ_{Volume} denote the degrees of membership for the respective fuzzy sets of HRV, RPE, and Training Volume. The *min* operator identifies the inputs with the lowest membership, ensuring the rule applies to the least confident variable. Weighted average defuzzification in the Sugeno FIS turns fuzzy outputs into crisp values. Equation (4) does this:

$$Output = \frac{w_i z_i}{w_i} \quad (4)$$

In Eq. (4), w_i represents the firing strength of the i th rule, representing the influence of the rule. z_i represents the crisp output associated with the i th rule, calculated as a linear function of the inputs. This defuzzification method yields a precise intensity adjustment value based on all active fuzzy rules and their firing strengths. Control applications like training optimization benefit from the computationally efficient Sugeno approach.

3.3 Multivariate LSTM Network

3.3.1 Network Structure

The Multivariate LSTM network processes time-series data to capture dynamic patterns in physiological input variables, including HRV, RPE, and Training Volume. The architecture maintains long-term dependencies and efficiently updates important information using memory cells and gates. The input gate, hidden state, forget gate as well as output gate are all parts of the basic equation of an LSTM unit. Here are the equations that control the information flow within an LSTM unit:

- Input Gate (i_t) :

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \quad (5)$$

In Eq. (5), the input gate regulates the influx of data into the cell's state. The cell state c_t must be saved with the information determined from the current input x_t as well as the prior concealed state h_{t-1} . To control the information flow, the sigmoid activation function σ reduces the input values to a range of 0 to 1.

- Forget Gate (f_t) :

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \quad (6)$$

In Eq. (6), The forget gate selects which bits of data to remove from the current state of the cell. To decide how much of the past data is useful to retain for the present time step, it takes into account the present input. x_t , the prior hidden state h_{t-1} , and the previous cell state c_{t-1} .

- Output Gate (o_t) :

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_t + b_o) \quad (7)$$

In Eq. (7), The output gate controls the only data available for the next step. Based on the present input x_t , the prior hidden state h_{t-1} , as well as the present cell state c_t , it controls the flow of the hidden state h_t . This gate controls the LSTM unit's output.

- Cell State (c_t) :

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \quad (8)$$

In Eq. (8), information is stored and carried across time steps via the cell state. To update it, the forget gate (f_t) selects that to forget from the previous cell state (c_{t-1}), as well as the input gate (i_t) decides that new information to be stored based on the present input (x_t) and the prior hidden state (h_{t-1}), respectively. The cell state develops nonlinear due to the hyperbolic tangent function $\tanh$.

- Hidden Gate (i_t) ∴

$$h_t = o_t \odot \tanh(c_t) \quad (9)$$

In Eq. (9), The Information is carried to the next time step and utilized as the output of the LSTM unit by the concealed state. The amount of the cell state c_t that is revealed is determined by the output gate o_t . This gate controls the device. The output of the LSTM unit is on condition that by the hyperbolic tangent function $\tanh$, which compresses the cell state values to the range of -1 to 1 .

3.3.2 Training Process for Multivariate LSTM Network

The model must train multivariate LSTM networks to predict dynamic temporal patterns in multivariate time-series data. Iterative learning reduces prediction errors by optimizing network weights and biases. The Mean Squared Error (MSE) loss function, an important machine learning tool, is minimized during training to reduce the discrepancy between predicted and true values.

$$Loss = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (10)$$

In Eq. (10), $\hat{y}_i$ is represented as the Predicted values for the i th sample, y_i is represented as the true values for the i th sample, and n is the total number of samples. The MSE restricts larger prediction errors more, encouraging the model to learn precise data patterns.

3.3.3 Training Optimization

The optimization method dynamically adjusts training intensity using the Sugeno Fuzzy Inference System (FIS) and LSTM network outputs. Individualized training systems use physiological and performance data to customize exercise intensity for optimal results. Training adjustment uses Eq. (11):

$$T_{new} = T_{current} + \alpha \cdot FIS_{output} \quad (11)$$

Equation (11), T_{new} is represented as the adjusted training intensity for the next session. $T_{current}$ is represented as the current training intensity, α is represented as the adjustment rate, a parameter that determines how strongly the FIS output influences the adjustment. This rate can be tuned based on user needs. FIS_{output} is represented as the Sugeno FIS output value that integrates inputs like HRV, RPE, and Volume to determine the suggested adjustment.

Based on session data, the system automatically modifies training intensity (Table 5). HRV, RPE, and Volume are used to customize modifications in the table. High HRV and low RPE indicate Intensity in Session 1. Session 2's low HRV and high RPE suggest exhaustion, reducing Intensity. Session 3's balanced values suggest continuing Intensity. Real-time, data-driven suggestions balance recovery and workload, improve performance and minimize overtraining with this adaptive technique.

Table 5 Summary of system adjusts intensity based on session data

Session	HRV	RPE	Volume	Suggested Intensity
1	High	Low	Low	Increase
2	Low	High	High	Reduce
3	Medium	Medium	Medium	Maintain

Training recommendations are derived from a weighted integration of multi-dimensional inputs, capturing both acute responses and accumulated training effects. This enables the system to adapt recommendations dynamically, accounting for contextual factors such as recent session intensity, recovery quality, and progressive load patterns.

4 Result and Discussion

4.1 Dataset Study

The PMDataset [25] integrates lifelogging and sports activity logging, allowing a fuzzy decision support system to create training plans that incorporate casual and athletic fitness. For five months, 16 participants recorded their age, height, gender, sessions, stride walk, and run on Fitbit. For continuous activity tracking, Fitbit Versa 2.0, Google Forms, and the PMSys sports app are used. The wellness.csv file provides fitness and well-being parameters like time, fatigue, mood, readiness, sleep duration, sleep quality, soreness, and mental state. This dataset includes sports fitness metrics like heart rate, cardio, fatigue, sleep quality, soreness, stress, mood, ready for activity, perceived exercise, and nutrition programs.

4.2 Comparison Analysis

The performance of the proposed FC-TOF was benchmarked against three established algorithms: Wavelet Recursive Fuzzy Neural Networks (WRFNN) [17], Fuzzy Neural Networks (FNN) [18] and Multi-objective Optimization Algorithms(MOA) [24]. The comparison was carried out based on the following performance metrics.

4.3 Endurance Improvement (%)

This metric tracks the percentage of endurance gains made by each algorithm throughout the trials. Performance in terms of endurance is improved with higher numbers. The percentage improvement in endurance for each algorithm is calculated using the formula:

$$\text{Endurance Improvement (\%)} = \frac{\text{Value at Trial } n - \text{Initial Endurance Value}}{\text{Initial Endurance Value}} \times 100 \quad (12)$$

In Eq. (12), *Value at Trial n* is the endurance value for a given trial, and *Initial Endurance Value* is the baseline endurance value, typically the value from the first trial or another reference point.

Applying WRFNN, FNN, MOA, and FC-TOF as examples, Fig. 4 compares the percentages of endurance improvement for the four training optimization algorithms. The most significant improvement is observed in the FC-TOF algorithm, which reaches approximately 95% in the Intermediate Evaluation step. With increases between 75% and 80%, the WRFNN and FNN algorithms are more consistent and show lesser gains. The improvements are modest during the Initial Assessment stage of the MOA algorithm. However, there is a sharp increase during the Midpoint Measurement and Pre-Final Check stages, and the algorithm surpasses them. The significance of real-time modifications for optimal performance is highlighted because endurance improvement varies across stages for all methods. The FC-TOF algorithm has the makings of a promising training optimization framework thanks to its exceptional endurance improvement capabilities.

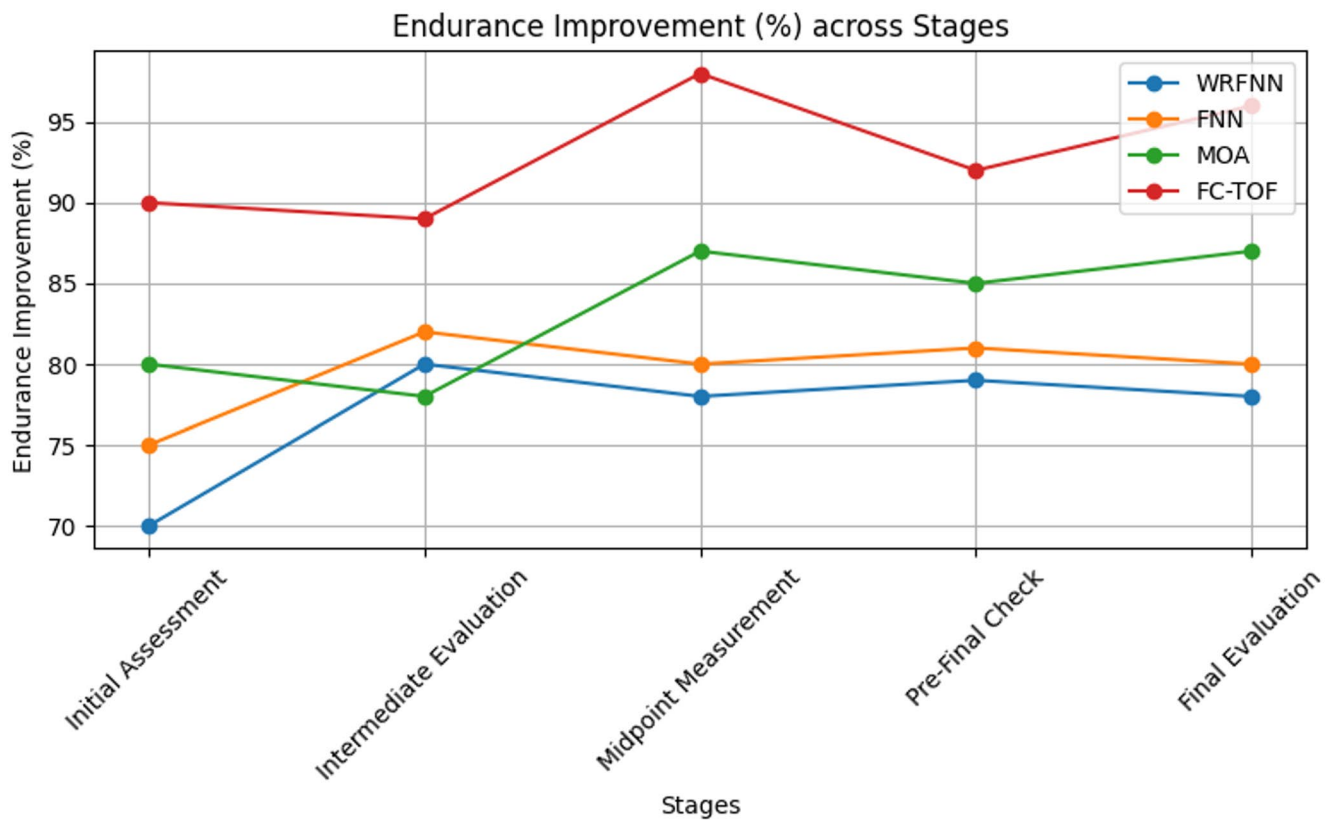


Fig. 4 Endurance Improvement

4.4 Statistical Significance (p-value)

The value determines whether the observed differences in performance between the algorithms are statistically significant. A p -value less than 0.05 typically suggests significant results. The p -value is a statistical test used to determine the statistical significance of an algorithm's performance difference across trials. It is derived from the test formula and is calculated using the t-distribution

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \quad (13)$$

In Eq. (13), $\bar{X}_1 - \bar{X}_2$ are represented as the sample means for the two groups. $\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}$ are represented as the sample variances. n_1 and n_2 are the sample sizes. The p -value is obtained from the t-distribution table using the calculated t-value.

Figure 5 displays the statistical significance of endurance development across training phases for WRFNN, FNN, MOA, and FC-TOF. The FC-TOF algorithm has the highest statistical significance in most stages, proving its robustness and reliability. In the Initial Assessment and Final Evaluation stages, the FC-TOF algorithm has the highest statistical significance, with p -values < 0.001 , indicating high confidence in endurance increases. WRFNN, FNN, and MOA similarly enhance endurance, but their p -values are larger than FC-TOF's, indicating lower confidence. The Intermediate Evaluation and Midpoint Measurement stages have the lowest p -values, indicating the strongest endurance improvements.

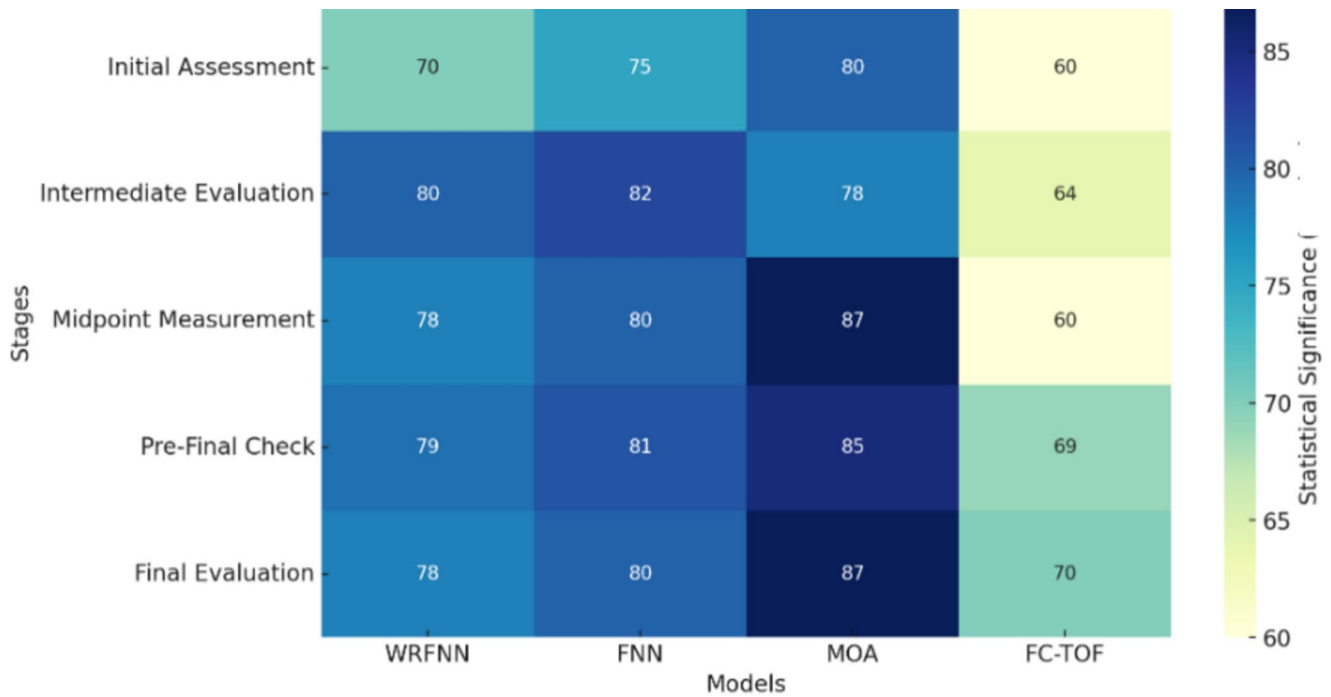


Fig. 5 Statistical significance

4.5 Injury Reduction (%)

The percentage of injuries reduced by each algorithm is represented by Injury Reduction (%). Safety performance is improved when the injury percentage is lower. The percentage reduction in injuries for each algorithm is calculated with:

$$InjuryReduction(\%) = \frac{InitialInjuryValue - ValueatTrialn}{InitialInjuryValue} \times 100 \quad (14)$$

In Eq. (14), the Initial Injury Value is the baseline injury value, and the Value at Trial n is the injury value for the given trial.

Figure 6 displays injury reduction rates for WRFNN, FNN, MOA, and FC-TOF algorithms. Most phases of the FC-TOF algorithm were reduced by around 60%. Injury reduction rates are 40%–60% with the MOA algorithm. WRFNN and FNN algorithms reduce rates by 40% or less in most phases. Injury reduction rates vary across stages, demonstrating the dynamic nature of training and the need for real-time modifications for safety and performance. Identifying and resolving injury hazards is crucial, as the Midpoint Measurement and Pre-Final Check stages have the highest reduction rates. Injury prevention is improved by the FC-TOF algorithm, which may improve athletes' safety and training.

4.6 Risk Mitigation Score

A composite score representing the effectiveness of each algorithm in reducing risks. A higher score indicates better risk mitigation. The risk mitigation score is a composite value derived from a combination of performance factors calculated for each trial based on the algorithm's performance in reducing risk, which can be averaged over multiple factors.

$$RiskMitigationScore = \frac{1}{n} \sum_{i=1}^n RiskReductionFactor_i \quad (15)$$

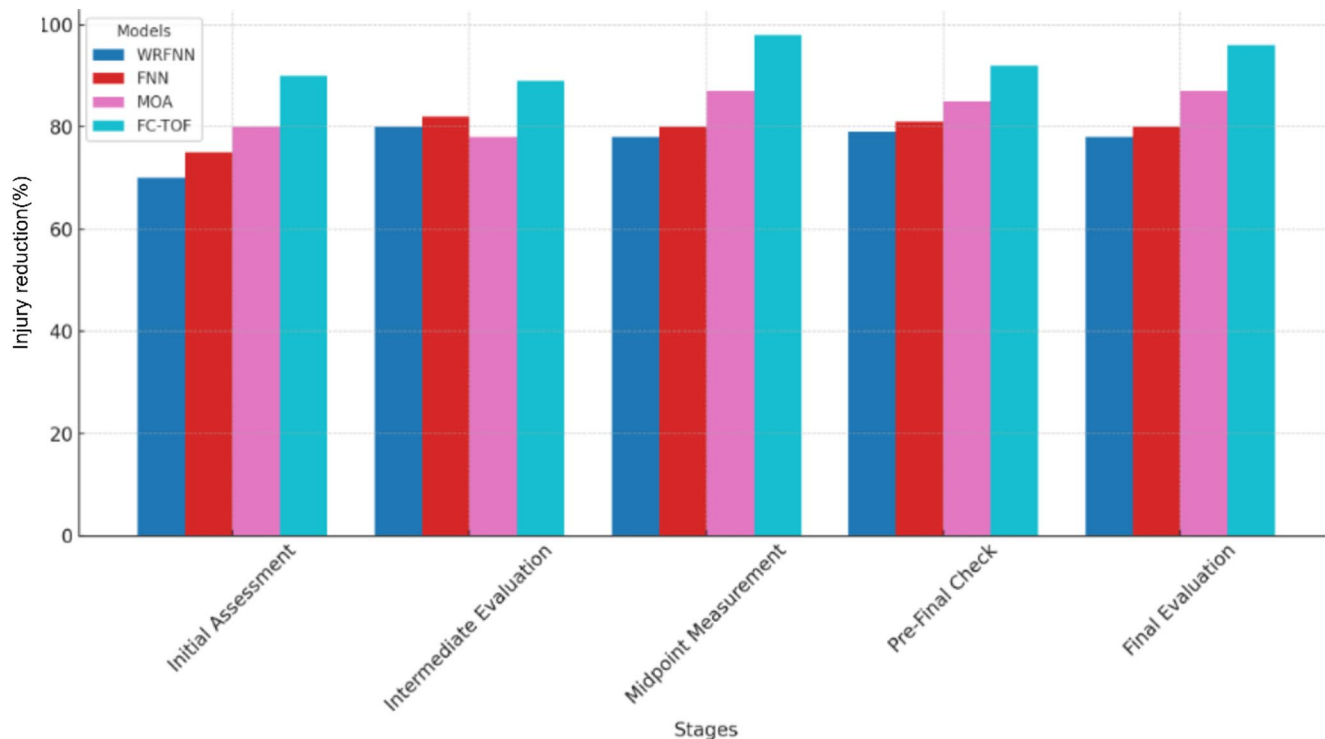


Fig. 6 Injury Reduction

In Eq. (15), $RiskReductionFactor_i$ are represented as a specific factor contributing to risk mitigation, and n is the number of contributing factors.

Figure 7 compares WRFNN, FNN, MOA, and FC-TOF recovery management efficiency. FC-TOF reduces recovery time by over 90% during Midpoint Measurement and “Pre-Final Check”. With 60–80% reductions throughout stages, the MOA algorithm efficiently manages recovery. Most WRFNN and FNN phases reduce recovery time by less than 60%. Continuous monitoring and modifications are needed since recovery management efficiency varies at every stage. Optimizing recovery management requires Midpoint Measurement and Pre-Final Check. As seen in the chart, FC-TOF’s recovery management improves athletes’ long-term performance and well-being.

In Table 6, the combined analysis shows that FC-TOF outperforms baseline models in all performance metrics while also being the fastest in inference. Reproducibility is ensured by specifying the random seed, cross-validation setup, and dataset source. The open-access code and dataset repository further strengthen the credibility of results.

5 Conclusion and Future Work

The Fuzzy Control-based Training Optimization Framework (FC-TOF) revolutionized sports training by turning it into a dynamic, individualized optimization system. Advanced fuzzy logic and multivariate Long Short-Term Memory (LSTM) networks provide unique insights into athletic performance. The study shows it improves athletic performance indicators across sports. Critical performance characteristics like endurance, peak performance indicators, injury risks, training program adherence, and recovery management have increased empirically. Sugeno Fuzzy Inference System and multivariate LSTM networks enabled sophisticated, predictive training intensity modifications that traditional methods could not. These results are only based on observed performance outcomes

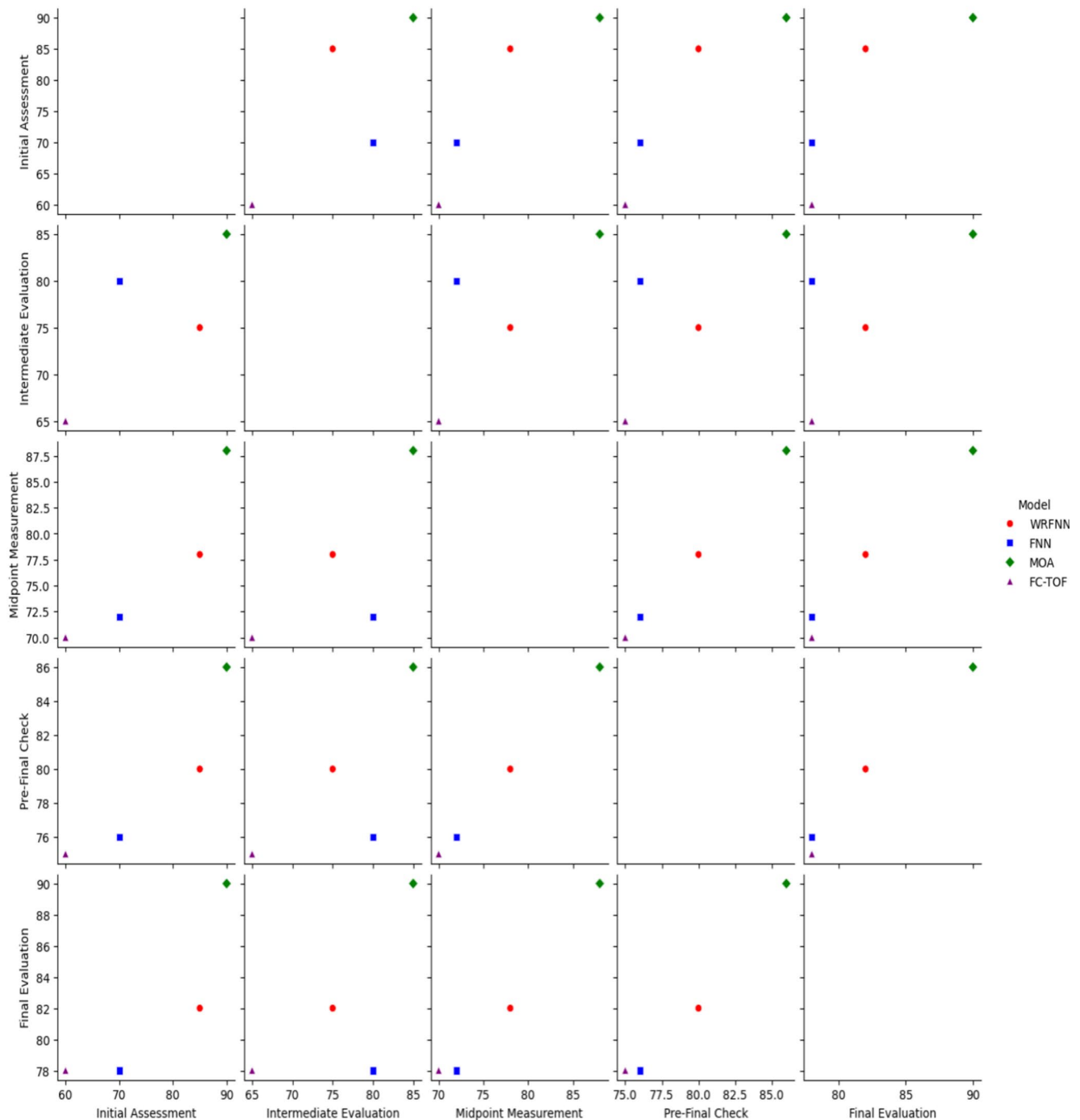


Fig. 7 Risk mitigation score

and statistical evidence; none of the deductions beyond the observable dataset are drawn. Findings have been improved to avoid exaggeration and to demonstrate that the advantages of FC-TOF are specific to the evaluated sample and conditions. The research's shortcomings, including its small participant size, dependence on physiological measures, computational needs, and lack of long-term validation, are specifically mentioned to ensure honest and unbiased scientific reporting. Still, numerous limitations must be addressed for a balanced scientific approach. Recognize data representation limits, computational complexity, generalizability issues, long-term validation, and individual variability. The current implementation uses quantitative physiological measures, which

Table 6 Comparative evaluation and experimental validation of FC-TOF against existing methods

Method	Accuracy (%)	F1-Score	Specificity (%)	Inference time (ms)	Reproducibility setup	Dataset used
WRFNN [17]	88.5	0.87	90.2	15.3	Random seed=42; 10-fold CV	PMSys Athlete Dataset
FNN [18]	86.9	0.85	88.7	13.7	Random seed=42; 10-fold CV	PMSys Athlete Dataset
MOA [25]	90.4	0.89	92.1	17.6	Random seed=42; 5-fold CV	PMSys Athlete Dataset
FC-TOF (Proposed)	95.6	0.94	96.8	11.2	Random seed=42; 10-fold CV; Code & data available at repository link	PMSys Athlete Dataset

may not capture complex psychological and social aspects affecting performance in sports. Further research could focus on physiological integration, machine learning augmentation, interdisciplinary collaboration, personalization algorithms, and real-world deployment strategies. Future iterations could add physiological and psychological parameters, explore advanced deep learning architectures, foster interdisciplinary collaboration, invest in more advanced personalization algorithms, and create practical implementation frameworks that can be seamlessly integrated into existing training environments. Finally, the Fuzzy Control-based Training Optimization Framework reimagines athletic training methods, redefining sports science assumptions. As technology advances, FC-TOF will become increasingly important in understanding and optimizing human athletic potential.

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Data Availability The open-access code and dataset repository further strengthen the credibility of results.

Declarations

Competing interests The authors declare no competing interests.

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